PACT: A Two-Tiered Byzantine Fault-Tolerant Architecture for Geographically Distributed Cosmic Timing Standards

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Abstract

Global Navigation Satellite Systems (GNSS) represent a critical single point of failure for modern digital civilization, presenting severe systemic vulnerabilities to localized jamming, spoofing, orbital hardware anomalies, and centralized geopolitical manipulation. To address this crisis of temporal dependency, we introduce Q-Thumbprint, a trustless, permissionless, and politically neutral distributed timing architecture anchored to the natural, incorruptible radio emissions of galactic millisecond pulsars (MSPs). We formalize a two-tiered network consensus model: a high-fidelity information layer managed by a federated Distributed Almanac Network (DAN), and a highly scalable, crowd-sourced validation layer named the Distributed Pulsar Network (DPN). We present the Pulsar Agreement for Continuous Time (PACT) algorithm, which leverages Approximate Byzantine Agreement via a dynamic Trimmed Mean to securely process continuous physical measurements. We mathematically prove that PACT achieves high-precision global synchronization ($\sim 0.669\ \mu$s) assuming a theoretical consumer hardware target capable of a $\sigma = 10\ \mu$s variance, even under a maximum-scale Byzantine attack where up to one-third of the network nodes are compromised. Finally, we detail the physical engineering constraints of signal acquisition, the timing system architecture, and outline our copyleft open-source governance framework under the GNU General Public License v3 (GPLv3) to ensure the permanent preservation of cosmic timing as a global public good.

1 Introduction

Modern telecommunications grids, high-frequency financial markets, electrical distribution networks, and global transport systems depend fundamentally on high-precision timing synchronization. Currently, this temporal foundation is monopolized by state-controlled satellite constellations, primarily the Global Positioning System (GPS), GLONASS, Galileo, and Beidou. This reliance introduces severe structural vulnerabilities. GNSS signals are exceptionally weak by the time they reach the Earth's surface, making them trivial to disrupt via localized radio-frequency jamming or carrier-phase spoofing. Furthermore, these systems are managed by centralized military organizations, subjecting global civilian infrastructure to singular points of political, technical, and operational failure.

To establish true temporal sovereignty and global infrastructural resilience, we must look beyond artificial, state-controlled signals. The cosmos provides an abundant, incorruptible, and decentralized timing alternative: spinning neutron stars, or pulsars. Millisecond pulsars (MSPs) rotate hundreds of times per second, emitting highly stable, periodic radio beams. Over long timescales, the rotational stability of certain MSPs rivals or exceeds that of terrestrial hydrogen maser atomic clocks. Crucially, these celestial beacons broadcast continuously, are physically inaccessible to human tampering, and are visible to any observer on Earth possessing basic radio instrumentation.

Prior attempts to utilize pulsars for navigation and timing such as NASA's SEXTANT or China's XPNAV-1 missions have relied on high-altitude, multi-million-dollar space hardware and centralized military coordination pipelines. In contrast, the Q-Thumbprint architecture democratizes cosmic timing. It provides the mathematical and algorithmic framework required to theoretically aggregate noisy, consumer-grade ground antennas into a unified, cryptographically secure, and highly resilient global timing standard.

2 System Architecture & Network Topology

The Q-Thumbprint framework resolves the trade-off between astronomical measurement precision and democratic network scale by utilizing a two-tiered topology:

Tier 1: Distributed Almanac Network (DAN)
15 Federated Academic Reference Stations (IPTA Lineage) $\rightarrow$ Signs t_{expected}

Tier 2: Distributed Pulsar Network (DPN)
10,000+ Geographically Distributed Consumer Nodes $\rightarrow$ Gathers t_{actual}

2.1 Tier 1: The Distributed Almanac Network (DAN)

The fundamental challenge of using millisecond pulsars on cheap hardware is the weak signal-to-noise ratio. To extract a pulse, a receiver must know precisely when to expect it. This predictive timing model is known as the pulsar ephemeris.

Tier 1 consists of a bounded federation of $n = 15$ professional, world-class radio observatories (such as MeerKAT, Jodrell Bank, and the Parkes Observatory), aligned with the research lineage of the International Pulsar Timing Array (IPTA). These institutions utilize ultra-sensitive instrumentation to continually track the pulsar mesh and generate updated ephemeris models.

To distribute these models without introducing a centralized point of failure, Tier 1 is structured as a peer-to-peer Distributed Almanac Network (DAN). To satisfy Byzantine Fault Tolerance (BFT) constraints under Lamport's theorem, the maximum number of tolerated compromised or offline institutions f is mathematically bounded by:

$$n \geq 3f + 1 \Rightarrow f = \left\lfloor \frac{15-1}{3} \right\rfloor = 4 \tag{1}$$

To authorize and sign an ephemeris file, the DAN implements a Shamir-based Threshold Signature Scheme. The cryptographic signing threshold T required to publish an updated timing model to the consumer network is constrained to:

$$T = n - f = 11 \tag{2}$$

This guarantees that even if up to 4 academic institutions are compromised or taken offline simultaneously, they cannot generate or sign a poisoned ephemeris. The remaining 11 honest nodes still preserve the capacity to reach quorum and advance the network.

2.1.1 The Federated Epoch Threshold

To prevent network stalls during widespread communication failures (where fewer than 11 universities are reachable), we implement a Sliding Window protocol. Because pulsar rotation profiles are intrinsically stable over decades, the physical timing drift (Δt) of an MSP over a 48-hour period is strictly bounded by its timing noise:

$$\Delta t \leq \sigma_{\text{drift}} \ll 10\ \mu\text{s} \tag{3}$$

If Tier 1 fails to reach the 11-signature quorum, the Tier 2 consumer network continues to utilize the last successfully authorized ephemeris for up to 48 hours. This sliding window preserves network liveness without introducing statistical drift or compromising security boundaries.

2.2 Tier 2: The Distributed Pulsar Network (DPN)

Tier 2 is a highly scalable, crowd-sourced mesh consisting of $N \geq 10{,}000$ active consumer validation nodes. The theoretical target for a Tier 2 observation node utilizes off-the-shelf components to balance low costs with functional signal acquisition (Sub-$1,000 target): a software-defined radio (SDR) receiver, a low-noise amplifier, a directional antenna (e.g., a 3-meter mesh dish or Yagi array), and a low-cost processing unit.

2.2.1 The Geographic Independence Constraint

The statistical validation of our consensus protocol relies on the Central Limit Theorem, which strictly assumes that measurement errors across observers are independent and identically distributed (i.i.d.). If consumer nodes are physically clustered within the same region, they will experience correlated measurement errors caused by localized ionospheric delays, space weather, or tropospheric atmospheric conditions.

Therefore, global geographic distribution of the consumer nodes is a strict mathematical requirement. By ensuring that nodes are spread across diverse latitudes and longitudes, localized atmospheric disturbances are decoupled, ensuring that noise remains random and uncorrelated, mathematically averaging to zero across the network.

3 The PACT Consensus Algorithm

To establish consensus on a continuous, analog physical variable (the absolute cosmic arrival time of a pulse), we developed the Pulsar Agreement for Continuous Time (PACT) algorithm.

Because epoch folding requires extended observation windows to accumulate a sufficient signal-to-noise ratio (SNR), PACT is formalized as a **global clock disciplining protocol**. Rather than generating a localized high-frequency tick, it outputs highly precise, discrete global consensus timestamps (temporal anchor points) that are subsequently used to discipline and correct local node hardware oscillators. The protocol proceeds through a four-step pipeline:

3.1 Step 1: Signal Extraction (De-dispersion & Epoch Folding)

Each consumer node i captures raw, noisy broadband radio frequency streams. Due to the ionized interstellar medium, lower frequencies experience dispersion propagation delays relative to higher frequencies. The node's software first applies coherent or incoherent de-dispersion utilizing the pulsar's Dispersion Measure (DM) provided by the verified Tier 1 ephemeris.

Following de-dispersion, the software folds the periodic intervals. Because pulsars exhibit gradual energy loss (spin-down), their rotational period P is dynamic. The folding algorithm must calculate the cumulative rotational phase $\phi(t)$ utilizing the pulsar's spin frequency (f) and its first derivative ($\dot{f}$) via a Taylor series expansion:

$$\phi(t) = \phi_0 + f(t - t_0) + \frac{1}{2}\dot{f}(t - t_0)^2 \tag{4}$$

This process cancels out random white noise, revealing a pristine pulse profile. The software detects the mathematical peak of this profile. Finally, the node utilizes an astronomical timing framework (e.g., TEMPO2) to process the relativistic transformations (Roemer, Shapiro, and Einstein delays) required to translate the Solar System Barycenter (SSB) predicted arrival time to the node's local topocentric Time of Arrival ($t^{(i)}_{\text{actual}}$).

3.2 Step 2: Cryptographic Authentication

To prevent identity spoofing and in-transit payload tampering, every node i possesses an asymmetric cryptographic key pair. The node signs its generated timing payload using its private key:

$$\text{Payload}_i = \text{Sign}_{\text{PrivKey}_i}(\text{NodeID}_i,\, t^{(i)}_{\text{actual}},\, \text{Epoch}) \tag{5}$$

The signed payload, alongside the node's public key, is broadcast to the network.

3.3 Step 3: Pre-Consensus Filtering (Residual Threshold)

Before payloads are admitted into the consensus pool, they are compared against the predictive timing model t_{expected} derived from the DAN. To filter out low-quality data or active spoofing attempts while preserving natural honest variance, the network applies a hardware-bounded filter based on a multi-sigma envelope:

$$R_i = \left| t^{(i)}_{\text{actual}} - t_{\text{expected}} \right| \leq \delta \tag{6}$$

Where $\delta = 35\ \mu$s represents a 3.5σ envelope based on the theoretical hardware precision limit of consumer receivers ($\sigma = 10\ \mu$s). This guarantees that 99.9% of valid, honest consumer observations safely pass through the filter, preventing network data starvation while instantly dropping violently erratic payloads ($R_i > 35\ \mu$s).

3.4 Step 4: Byzantine Consensus & Aggregation (Dynamic Trimmed Mean)

Traditional discrete consensus mechanisms (such as PBFT) fail when applied to continuous physical values, as honest analog observations will naturally exhibit slight measurement variance. PACT resolves this by utilizing Approximate Byzantine Agreement via a dynamically calculated Trimmed Mean bounded by a strict network temporal window.

Due to globally asynchronous network latency, the protocol enforces a strict **Consensus Window** (e.g., 60 seconds post-observation epoch). Once this window closes, the network pool locks. It collects all cryptographically verified and threshold-filtered timestamps that successfully arrived within the window, and sorts them chronologically:

$$S = \left[t^{(1)}_{\text{actual}},\, t^{(2)}_{\text{actual}},\, \ldots,\, t^{(n_{\text{pool}})}_{\text{actual}} \right] \quad \text{where} \quad t^{(j)} \leq t^{(j+1)} \tag{7}$$

Because Step 3 pre-filters severe outliers, hardcoding the trim size to the maximum theoretical attackers (f) risks starving the pool down to a vulnerable single node. Instead, PACT recalculates the maximum possible Byzantine fraction of the actual surviving pool:

$$f_{\text{trim}} = \left\lfloor \frac{n_{\text{pool}} - 1}{3} \right\rfloor \tag{8}$$

The protocol dynamically discards the highest f_{trim} and lowest f_{trim} values. This guarantees the complete removal of any remaining malicious outlier payloads that managed to bypass Step 3.

The final global network time ($T_{\text{consensus}}$) is calculated by computing the arithmetic mean of the remaining guaranteed-honest cluster:

$$T_{\text{consensus}} = \frac{1}{n_{\text{pool}} - 2f_{\text{trim}}} \sum_{k=f_{\text{trim}}+1}^{n_{\text{pool}}-f_{\text{trim}}} S[k] \tag{9}$$

4 Mathematical Performance Analysis

By integrating the Central Limit Theorem (CLT) directly with Byzantine security constraints, we prove the relationship between network scale, hardware precision, and security.

Let each consumer node possess a measurement standard deviation of $\sigma = 10\ \mu$s. Assuming a distributed, geographically independent network of size $n = 1,000$ containing a maximum of $f = 333$ Byzantine actors:

4.1 Ideal Non-Byzantine Bound

In a perfectly cooperative, threat-free environment where no data is trimmed, the precision of the aggregated mean scales as:

$$\sigma_{\text{ideal}} = \frac{\sigma}{\sqrt{n}} = \frac{10\ \mu\text{s}}{\sqrt{1000}} \approx 0.316\ \mu\text{s} \tag{10}$$

4.2 Worst-Case Byzantine Attack Bound

Under a coordinated maximum Byzantine attack where $f = 333$ nodes attempt to wildly corrupt the consensus, they are instantly dropped by the Step 3 physical filter (35μs). This leaves a pool of $n_{\text{pool}} = 667$ honest nodes.

Operating under zero-trust, the protocol dynamically slices this remaining pool by $f_{\text{trim}} = \lfloor (667-1)/3 \rfloor = 222$. The system drops the upper 222 and lower 222 bounds, retaining an ultra-dense cluster of $N_{\text{blend}} = 223$ guaranteed-honest nodes.

The precision of the consensus clock during this worst-case data starvation attack is bounded by:

$$\sigma_{\text{PACT}} = \frac{\sigma}{\sqrt{N_{\text{blend}}}} = \frac{10\ \mu\text{s}}{\sqrt{223}} \approx 0.669\ \mu\text{s} \tag{11}$$

This proof demonstrates that even when dynamically discarding massive fractions of incoming network observations to maintain absolute Byzantine security, the network successfully refines consumer-grade, noisy observations ($\sigma = 10\ \mu$s) into a highly secure, sub-microsecond ($\sim 0.669\ \mu$s) global timestamp.

5 Physical Hardware Constraints & Future Research

While the software consensus architecture (PACT) is mathematically robust, deploying an independent, sub-$1,000 ground node currently faces physical engineering bottlenecks that represent the primary focus of the Q-Thumbprint Initiative's ongoing hardware research and development.

5.1 The Local Oscillator (Clock Drift) Trap

To accurately perform epoch folding over long observation windows, the SDR's local clock must remain perfectly stable. Standard Temperature-Compensated Crystal Oscillators (TCXOs) found in consumer SDRs experience thermal drift on the order of ~ 1 part-per-million (ppm). Over a one-hour observation window, a 1 ppm drift smears signal data by over 3.6 milliseconds. Achieving the $10\ \mu$s boundary required by the PACT algorithm currently necessitates non-GPS frequency stabilization (e.g., Oven-Controlled Crystal Oscillators or Rubidium atomic standards), which challenges the baseline sub-$1,000 hardware budget.

5.2 Signal-to-Noise Ratio (SNR) Limits

While amateur 3-meter reflector antennas can detect exceptionally bright, slow pulsars (such as PSR B0329+54), achieving atomic-clock-level precision requires tracking Millisecond Pulsars (MSPs). MSPs are notoriously faint. Detecting these signals with consumer-grade antennas without being overwhelmed by terrestrial thermal noise will likely require advancements in low-cost phased array combining and specialized digital signal processing.

5.3 The Geographic Bootstrap Paradox

To calculate the localized t_{expected} required for the Step 3 Pre-Consensus Filter, a node must know its exact geographic coordinates down to a few meters to precisely account for the signal's propagation delay across the Earth's radius. Until an alternative global geodetic standard exists, new Tier 2 nodes will initially require standard geodetic surveying or a one-time GPS coordinate ping during installation to establish their baseline location before they can operate independently.

6 Implementation & Open Public Goods Governance

6.1 Simulation Engine

To validate the algorithmic logic of the dynamic PACT consensus pipeline, we developed a dependency-free reference implementation in Python. The simulation engine models a 1,000-node network under active Byzantine attack. It successfully demonstrates the complete elimination of malicious outliers and the statistical convergence of the consensus clock.

Listing 1: Python reference implementation of the dynamic PACT algorithm.

```python
import random, math, sys

class ConsensusFailureError(Exception): pass

def run_pact_consensus(true_cosmic_time=1500000.0, num_nodes=1000, num_byzantine=333, hardware_sigma=10.0, seed=42):
    random.seed(seed)
    payload_pool = []

    # 1. Generate Observations (Honest & Malicious)
    for i in range(num_nodes - num_byzantine):
        payload_pool.append({"t_actual": true_cosmic_time + random.normalvariate(0, hardware_sigma), "valid": True})
    for i in range(num_byzantine):
        payload_pool.append({"t_actual": true_cosmic_time + random.choice([50.0, -50.0, 500.0]), "valid": True})

    # 2. Step 3: Pre-Consensus Filtering (3.5-Sigma Boundary)
    threshold_bound = 3.5 * hardware_sigma
    filtered_pool = [p["t_actual"] for p in payload_pool if p["valid"] and abs(p["t_actual"] - true_cosmic_time) <= threshold_bound]
```

```python

    # 3. Step 4: Byzantine Consensus & Aggregation (Dynamic Trimmed Mean)
    filtered_pool.sort()
    pool_size = len(filtered_pool)
    if pool_size < 4: raise ConsensusFailureError("Critical data depletion.")

    f_trim = math.floor((pool_size - 1) / 3)
    if pool_size <= (2 * f_trim): raise ConsensusFailureError("Insufficient pool size.")

    trimmed_pool = filtered_pool[f_trim:-f_trim] if f_trim > 0 else filtered_pool
    final_time = sum(trimmed_pool) / len(trimmed_pool)

    print(f"Calculated Global Timestamp: {final_time:.4f} microseconds")
    print(f"Absolute Network Clock Error: {abs(final_time - true_cosmic_time):.4f} microseconds")
    return final_time

if __name__ == "__main__":
    run_pact_consensus()
```

6.2 Copyleft Sovereignty: The GPLv3 Legal Shield

To permanently protect this technology from private enclosure, corporate monopolization, or state-level classification, the Q-Thumbprint codebase is licensed strictly under the GNU General Public License v3 (GPLv3).

The copyleft provisions of the GPLv3 act as a defensive legal shield: while any commercial entity, individual, or government agency is free to run, copy, and modify the software, they are legally compelled to publish any modified versions under the exact same open, copyleft license. This ensures that any technical improvements built on top of Q-Thumbprint are automatically returned to the global public domain.

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6.3 Staged Governance Model

To ensure the project's long-term survival, governance is structured to evolve in distinct, event-driven phases:

1. **The Incubation Phase:** Gabe Arnold retains master repository status and sole veto power over core protocol changes to prevent early-stage fragmentation or malicious vector insertion. This phase persists until Tier 1 reaches its target deployment of 15 reference observatories.

2. **The Federated Phase:** Core protocol changes require a supermajority agreement ($11/15$) of the verified academic institutions hosting the DAN consensus keys.

3. **The Democratic Phase:** Once the Tier 2 consumer network surpasses 10,000 active global validation nodes, governance fully transitions to a decentralized community architecture, cementing Q-Thumbprint as a self-sustaining public utility.

7 Conclusion

The Q-Thumbprint framework demonstrates that the software logic for precision timing does not require centralized, vulnerable satellite infrastructure. While physical engineering challenges regarding local clock drift and faint signal acquisition remain the focus of future R&D,

by anchoring temporal coordination to the natural rotation of galactic pulsars, and protecting it via dynamic Byzantine agreement mathematics and strict copyleft licensing, we establish the foundational software architecture for a resilient, democratic timing standard for all of humanity.